RREC
Powering a Green Future

DISTRIBUTION UTILITY MEET

DISTRIBUTION UTILITY MEET 2026 ● SESSION 3

Role of RE in Managing New and Emerging Loads and Ensuring Supply Reliability and Power Quality

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What we will cover

01 **Before**
Traditional Loads

02 **Now**
New and Emerging Loads

03 **How RE’s Role is Evolving**
Energy, Firm, Flexible and Stable

04 **Managing the Loads**
Matching each Load to RE, Reliability, and Power Quality

01

Before

Traditional Loads

Before | **Now** | **RE's Role** | **Managing**

Yesterday's grid served familiar, predictable loads

HOUSEHOLDS - 21% OF SALES

Lights, fans, TV and a little cooling

- One evening peak, around dusk
- Steady growth, easy to forecast

AGRICULTURE - 40% OF SALES

Irrigation pumps on rural feeders

- About 9% of connections, 40% of sales
- Peaks on winter mornings (Rabi)

INDUSTRY - 27% OF SALES

Factories and mills buying from the DISCOM

- Steady daytime baseload

COMMERCIAL - 7% OF SALES

Shops, offices and hotels

- Daytime and early evening
- Predictable with the seasons

PUBLIC LOADS - UNDER 5%

Railways, water works, streetlights

- Small and steady
- Rarely a planning problem

PLANNING

One demand curve, planned years ahead

- Growth of about 5% a year
- Served by long coal and hydro contracts

Shares of Rajasthan DISCOM sales, FY2023-24, RERC-approved; the mix has barely moved.

Sources: IIT Kanpur ERC Hub (RERC sales), Ministry of Power, 2016, Copenhagen Consensus / IL&FS, India Climate & Energy Dashboard

02

Now

New loads are arriving together, and each one uses power at a different time of day

Before | Now | RE's Role | Managing

New loads behave differently from the old ones

BEFORE		NOW
Predictable and slow Daily and seasonal curves repeated year after year	→	**Fast: demand can jump within a second** AI Data Centre loads swing by more than 50%
Spread across millions Households and farms: no single customer mattered	→	**Concentrated: 100 to 1,000 MW campuses** One data centre can equal a small city on one feeder
One-way flow Power moved from the grid to the consumer	→	**Two-way: rooftop, batteries, V2G** 2.65 lakh rooftop systems in Rajasthan already export power
Planned years ahead Plants took years; load grew slowly	→	**Built in months, faster than the grid** Chargers and data centres outpace grid upgrades

In Rajasthan, peak demand has grown 79%: from 10,961 MW (FY2015-16) to 19,617 MW (FY 2025-26).

Sources: Ministry of Power, 2016, CEA, January 2026, IEA, India Climate & Energy Dashboard

New loads, and when each one uses power

AI data centres

26.3 GW

more by 2031-32

Flat, 24×7, with fast swings

Electric vehicles

8.3%

of car sales, Sept 2026

Evening and night charging

Green hydrogen

3,000 MW

electrolyser capacity a year awarded

Farm pumps

5,535 MW

PM-KUSUM installed, No. 1 in India

Rooftop solar

3.48 lakh

homes with PM Surya Ghar, 4th in India

Batteries

6.0 GWh

of batteries already operating in Rajasthan

Sources: Business Standard (MoP), Autocar Pro (Vahan), ETV Bharat (NGHM), Renewable Mirror (KUSUM), pv magazine India (Surya Ghar)

Data centers and AI agents: the biggest, flattest load

Data centers, and the AI agents and robots they serve, need steady power all day, every day.

SIZE

26.3 GW

more AI data-centre load projected by 2031-32

- About 1.6 GW operates in India today
- Campuses run 100 to 1,000 MW each
- Rajasthan: over ₹43,000 crore of proposals

WHEN IT USES POWER

Flat, 24 hours a day, every day

- Cooling adds an afternoon peak on top of the base load
- It cannot wait for the sun, so there is no cheaper hour to move it to

RISK: TO RELIABILITY AND POWER QUALITY

Fast swings, and no tolerance for poor power quality

- AI load can jump by more than 50% within a second
- Campuses are built in months; grid upgrades take years

What RE can do - Firm, round-the-clock RE with a battery on site

- A battery that steadies the grid (grid-forming) absorbs the jumps
- Rajasthan's policy offers banking, transmission, wheeling exemptions
- Virtual PPAs

Sources: Business Standard (MoP), IEA · pv magazine (SECI), The Hawk (Rajasthan)

EVs and small many loads: shiftable, if priced right

Lakhs of small loads, like EVs, pumps and rooftop systems, sit on feeders and act together like one big load.

SIZE

8.3%

of passenger-car sales in September 2026, a record 35,948 e-cars

- FY26: 24.5 lakh EVs sold in India
- Rajasthan: about 4.2 lakh EVs and 1,515 public stations
- Tendered for 696 more public stations

WHEN IT USES POWER

Evening and night at home; depots in concentrated blocks

- Two- and three-wheelers plug in around 7 to 8 pm, as solar fades
- Fleet and bus depots draw large loads at a single point

RISK: TO RELIABILITY AND POWER QUALITY

Adds to the evening peak and strains local equipment

- Local transformers overload when many cars charge together
- Chargers can distort the power supply (harmonics)

What RE can do - Move charging into solar hours and add storage at hubs

- Solar-hour time-of-day tariffs give owners a reason to charge by day
- Solar-plus-battery charging hubs for fleets and highways
- V2G (EVs sending power back): parked cars can help at the evening peak

Sources: Autocar Pro (Vahan), Mercom (RREC tender), The Wire (ISGF V2G)

V2G and VPPs

V2G (Vehicle-to-Grid)

A parked EV sends power back to the grid, for example at the evening peak, and charges again by day.

In Rajasthan

- RICEP 2024 supports EV charging stations run on RE
- Base today: about 4.2 lakh EVs and 1,515 public stations
- Missing: Bidirectional chargers, rules for exporting & metering power, and a price that makes it worth an owner's while

In India

- CEA (2023): about 500 EVs together can supply 1 to 2 MW
- ISGF ran a small pilot with four EVs across utilities in Delhi and Kerala

VPPs (Virtual Power Plant)

Many small batteries, solar systems and flexible loads run together as one plant and offered to the grid as one unit.

In Rajasthan

- **RICEP 2024,**
  - Promote solar projects based on Virtual Power Purchase Agreements (VPPAs)

In India

- No national VPP rules exist yet

Sources: RICEP 2024, Down To Earth, Mobility House, energy-storage.news, CEA V2G report, ISGF demo

Green hydrogen: the most flexible new load

Electrolysers use power to split water into hydrogen.

SIZE

3,000 MW

a year of electrolyser manufacturing awarded under the National Green Hydrogen Mission

- Production awards: 7.56 lakh tonnes a year (Aug 2026)
- Commissioned output is still very small
- Rajasthan policy target: 2,000 KTPA by 2030

WHEN IT USES POWER

Can follow solar and wind output within seconds to minutes

- Can ramp up at midday and ease down in the evening
- Downstream ammonia plants prefer a steady feed

RISK: TO RELIABILITY AND POWER QUALITY

Big blocks of demand at one point, which can distort power quality

- Power converters distort the supply and need extra voltage support

What RE can do - Co-locate with solar and wind; use as a flexible sink

- Sit electrolysers beside RE parks and avoid long transmission
- Run at full power in surplus hours; back off at night
- Rajasthan policy: 50% transmission and distribution rebate for early projects

Sources: ETV Bharat (NGHM), Business Standard (spend to date), GH2 (state policy)

Farm pumps: from night supply to daytime solar

Irrigation pumps are Rajasthan's largest customer group, and the easiest to move into the sun.

SIZE

40%

of Rajasthan DISCOM sales go to agriculture, from about 9% of connections

- About 1.9 million agricultural connections
- PM-KUSUM: 5,535 MW installed, No. 1 in India
- Daytime farm supply now reaches 26 districts

WHEN IT USES POWER

Winter mornings in the rabi season

- Wheat and mustard irrigation peaks from December to February

RISK: TO RELIABILITY AND POWER QUALITY

Night supply, a large subsidy, and weak rural voltage

- Many farm feeders have been supplied at night
- Focus is on shifting to daytime supply via KUSUM

What RE can do - solar on the feeder, beside the pumps

- PM-KUSUM Component C solarises whole agricultural feeders
- Daytime supply cuts the cost of buying power for the night

Sources: Renewable Mirror (KUSUM), IIT Kanpur ERC Hub (RERC sales), CEIC (consumers), Down To Earth

Rooftop solar: customers become producers

Households and businesses generate power by day and still rely on the grid at night.

SIZE

3.48 lakh

households with rooftop solar under PM Surya Ghar in Rajasthan, 4th in India

- 3,548 MW total rooftop solar capacity (July 2026)
- 1,360+ MW capacity added under PMSG
- India: about 50 lakh households and 14.8 GW (July 2026)

WHEN IT USES POWER

Sends power to the grid at midday, draws it back in the evening

- Demand from the grid drops sharply in sunshine hours
- It rises quickly again as the sun sets

RISK: TO RELIABILITY AND POWER QUALITY

Power flowing back, high voltage, and a steeper evening rise

- Power flowing back overloads low-voltage feeders and raises voltage
- DISCOM daytime sales fall while evening needs grow

What RE can do - Smart inverters, local batteries, virtual net metering

- Smart inverters hold voltage steady
- Batteries at distribution transformers store midday exports
- Virtual net metering: one plant serves many government buildings

Sources: pv magazine India (Surya Ghar), Down To Earth

Batteries: a load that charges at noon

Storage is also a customer: it charges when solar is in surplus and returns the power after sunset.

SIZE

6.0 GWh

of batteries already operating in Rajasthan, about 43% of India's 14 GWh

- About 25.4 GWh more in the pipeline
- About 6 GWh of standalone batteries awarded by RVUNL
- Standalone prices: ₹1.775 to ₹2.85 lakh per MW a month

WHEN IT USES POWER

Charges in solar hours, discharges after sunset

- Soaks up midday surplus that would otherwise be curtailed
- Supplies power in the hours after sunset

RISK: TO RELIABILITY AND POWER QUALITY

Low risk: the one new load that helps the grid

- About 15% is lost in a charge-discharge cycle (85% efficiency)
- Needs revenue beyond tariffs: an ancillary-services market; value stacking is pending

What RE can do - Place batteries where solar is stuck and feeders are weak

- Standalone batteries at 220 kV grid points where solar is held back
- Small batteries at feeders for EV and local peaks
- Merchant batteries: 100% wheeling waiver for 7 years on peak supply

Sources: pv magazine India (RVUNL), Energetica India (DISCOM pilot), PSU Watch (Pugal)

03

How RE’s role evolves

From supplying energy to supplying firm, flexible and stable power.

Before | **Now** | **RE’s Role** | **Managing**

RE has taken four jobs

1

Energy

Supplied cheap units; the grid adjusted

- Renewable purchase obligations and must-run status
- Bhadla solar at ₹2.44 per unit, energy only (2017)

2

Firm

Supplied power at chosen hours

- SECI peak-power tender: weighted ₹4.04/kwh (2020)
- First FDRE auction: ₹4.38/kwh (Nov 2023)
- Pugal solar with 4-hour storage: ₹3.42/kwh (2026)

3

Flexible

Helped demand move into the sun

- ToD tariffs; aggregators can pool batteries

4

Stable

Supports the grid in the first seconds

- CEA: Batteries of 50 MW or more to steady the grid (grid-forming) and restart it after a blackout (black start)

The answer to new loads is a more capable RE

Sources: pv magazine (2020 tender), IEEFA (first FDRE), Energetica (RERC DF/DSM), Renewable Watch (CEA standards)

04

Managing the loads

Reliability and Power quality: Matching each load to the RE that suits it.

Before | Now | RE's Role | Managing

Smart managing of demand and supply

Shift demand

Demand side

Help customers use power when RE is plentiful

- Time-of-day tariffs, including for EV charging
- Daytime farm supply on solar feeders (PM-KUSUM)

Make supply firm

Supply side

- Solar with four hours of storage, as at Pugal (₹3.42/kWh)
- Round-the-clock RE: ₹5.25 a unit (SECI, August 2026)
- Standalone batteries at grid points where solar is held back

Enable more projects

Enabler Side

- DRE-PMT: Management to Monitoring
- RE Developer Connect Portal: Ease of Doing Business (register and seek connectivity online)
- Government Building Solarization use of virtual/group net metering, shared solar, up to 1 MW
- Wind repowering, as RRECL is planning it for its own wind assets

Sources: pv magazine (round-the-clock tender), Down To Earth (flexibility rules), The Wire (ISGF V2G demonstration), Renewable Mirror (PM-KUSUM)

Match each new load to the RE that suits its

Steady Loads

Data centers

Steady, high, and sensitive to power quality

- Round-the-clock RE
- An on-site grid-forming battery for fast swings
- Group-captive supply under open access

Evening and night

EVs

- Four-hour storage, so solar lasts into the night
- Solar-hour ToD tariffs move charging to the day
- Solar-plus-battery charging hubs

Move to the sun

Hydrogen, farm pumps, rooftop, batteries

- Electrolysers and batteries soak up midday surplus

Sources: pv magazine (round-the-clock tender), Down To Earth (flexibility rules), The Wire (ISGF V2G demonstration)

Supply-side response and resource adequacy

PLAN

Put new loads into the state plan

- Rajasthan adequacy rules: keep at least 10% extra capacity in reserve
- Add data-centre, EV and C&I forecasts
- The STU prepares one consolidated storage plan
- Power system planning in close coordination with transport, industry and digital infrastructure development planning

CONTRACT

Buy firm power, not just energy

- Solar with 4-hour storage, sold as a firm product
- Specify availability, not only units

PLACE

Put batteries where solar is stuck

- About 4 GW of RE faces peak-hour restrictions
- Spare transmission is shrinking in FY27
- Standalone batteries at 220 kV nodes are fastest
- Planning of BESS at Bhadla park for utilization of 680 MW available grid connectivity during non-solar hours.

Sources: SolarQuarter (RERC adequacy), pv magazine (SECI), BW Businessworld (curtailment)

Rajasthan is building storage four ways

With solar

6,400 MWh

at Pugal

Solar with four hours of storage after sunset

₹3.42 per unit

Standalone, on the grid

6 GWh

awarded

₹1.775 to ₹2.85 lakh per MW a month

At the feeder

688 MWh

DISCOM pilot

172 MW of small units at 33/11 kV substations

Built for local peaks

Pumped storage

900 MW

5,400 MWh at Rampura

6 hours a day; 40 years, BOOT

Tender in-progress.

Sources: pv magazine India, Energetica India, RESI India (GEOA 2025), PSU Watch

Thank you